

Advanced Prompt Engineering Techniques

Foundational Strategies Prompt Engineering is the practice of structuring inputs to LLMs to elicit desired outputs. Standard strategies include Zero-Shot prompting (asking for response with no examples) and Few-Shot prompting (providing input-output examples).

Advanced Reasoning Frameworks - Chain-of-Thought (CoT): Encourages the model to output intermediate reasoning steps before the final answer, improving math and logic reasoning. - ReAct (Reason + Act): Combines reasoning traces and action steps, enabling the model to search external databases or use APIs and evaluate results before generating replies. - Tree of Thoughts (ToT): Extends CoT by exploring multiple reasoning paths (branches) and self-evaluating choices at each step.

Security Concerns Prompt Injection occurs when adversarial users input text that overrides system instructions, forcing the model to perform unauthorized actions or leak sensitive developer instructions.